

MFE 230GA · FINAL PROJECT · FALL 2026

Follow the Workers

Do labor-market flows predict which U.S. industries outperform?

TEAM

Romain ALMEIDA
Elouan BAHRI
Al Yazid BENSaid
Piero PELOSI
Alex ROESLER

COURSE

MFE 230GA – Active Equity Management

INSTRUCTORS

Raffaele Savi and Gerald Garvey
BlackRock

GRADUATE STUDENT INSTRUCTOR

Vinicio DeSola

1 Executive summary

Can public labor-market flows predict which of 13 U.S. industry groups outperform? We rank the groups each month on the year-over-year change in job openings, hires, quits, layoffs, hours and wages, as known at the time, and hold a market-neutral book long the top three and short the bottom three. The design was preregistered, frozen and opened once on 142 untouched months (November 2014 to August 2026). The primary strategy lost 4.6% a year after costs (Sharpe -0.58); its forecasts were uncorrelated with returns (information coefficient, IC, of -0.021 , $t = -1.08$) and a placebo test could not tell it from noise ($p = 0.72$). We conclude: *Do not implement*. It failed because the preregistered signs of layoffs and hours were wrong, the one stable one-month signal (wage growth) was dropped during the build, and the labor market never tightened in the research years, so the tightness hypothesis could not be tested. Across horizons the data are consistent with a slower, opposite effect: industries that hire and lose workers fastest earn lower returns over the following 9 to 24 months, which investment-based asset pricing would predict. A 12-month book on that sign is positive in every window (test Sharpe $+0.29$) but is post hoc, lost money in the first half of the test and does not survive deflation. It runs as paper books in a forward test frozen on 3 October 2026, with no capital. [TODO: Al Yazid: one sentence on Claude as research assistant, in the team's words]

2 What did we try?

2.1 Idea: three hypotheses about labor flows and industry returns

Our thesis is that the labor market reports an industry's prospects before its earnings do. The Bureau of Labor Statistics (BLS) publishes job openings, hires, quits and layoffs by industry every month in the Job Openings and Labor Turnover Survey (JOLTS), and hours and earnings in the Current Employment Statistics (CES). The series are free and timely, and industry rotation rarely uses them; it relies on price momentum (Moskowitz and Grinblatt, 1999; Zarattini and Antonacci, 2024). We state three hypotheses (Figure 1).

- *H1, demand channel* (preregistered and tested). Rising openings, hires and hours signal growing demand for an industry's output before it reaches reported sales, so the industry outperforms over the next one to three months (Hong et al., 2007).
- *H2, cost and investment channel* (tested post hoc). Firms that hire heavily convert growth options into assets in place and pay adjustment costs, so their expected returns fall over the following 6 to 24 months (Belo et al., 2014); wage pressure squeezes margins, and labor-market frictions make labor-intensive firms riskier (Kuehn et al., 2017; Donangelo et al., 2019).
- *H3, tightness moderator*. When vacancies outnumber the unemployed, hiring is expensive and H2 should dominate H1. We measure tightness by $T_t = \log(\text{job openings}_t / \text{unemployed}_t)$; $T > 0$ means more vacancies than unemployed ($V/U > 1$).

The preregistered strategy traded H1 at a one-month horizon. H2 was examined only after the test had been opened and carries the post-hoc label wherever it appears.

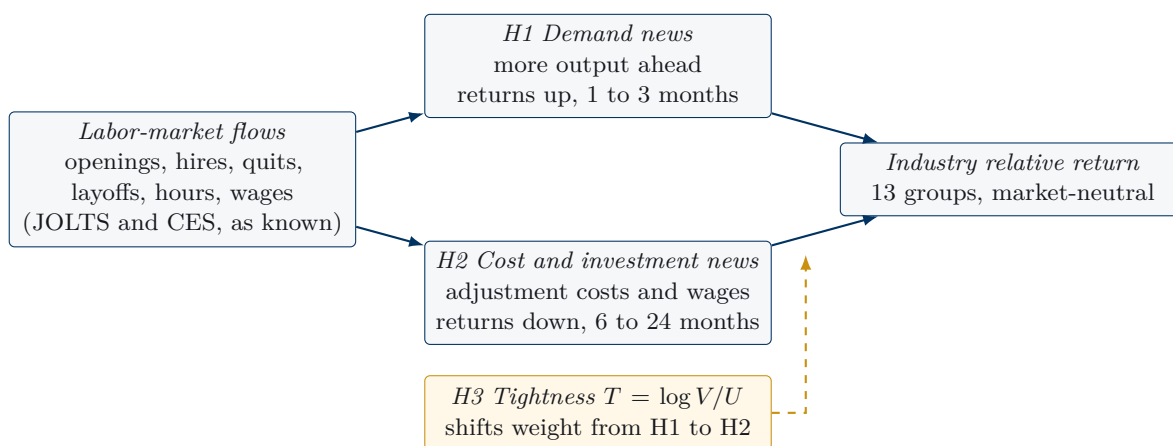


FIGURE 1. The three hypotheses. Labor flows carry two kinds of industry news with opposite sign and different horizons (H1, H2); tightness should change which one dominates (H3).

Table 1 lists each signal, the sign we preregistered under H1 and the mechanism behind it. The last column, added after the fact, shows what the data say.

TABLE 1. Signals, preregistered signs and what the data say. CONFIRMATORY (design) / POST-HOC (last column)

Signal	Definition (group g , month t)	Prior	Mechanism	IC, $h=1$
F1 openings	12-month change in 3-month average openings rate	+	H1 demand news	−0.005
F2 hires	same, hires rate	+	H1 demand news; H2 at long horizons	+0.013
F3 quits	same, quits rate	none	H2, worker bargaining power	+0.009
F4 layoffs	same, layoffs rate	−	H1, negative demand shock	+0.028
F5 hours	12-month log change, average weekly hours	+	H1, intensive margin	−0.010
W wages	12-month log change, average hourly earnings	dropped	H2 cost pressure (−) vs. labor risk premium (+)	+0.039

IC (information coefficient) is the average monthly Spearman correlation between the signal and the next month's industry relative return, May 2004 to August 2026, revised data with publication lags. Throughout, t -statistics use Newey–West standard errors, which allow for autocorrelation (HAC); only W has $|t| > 2$ ($t = 2.50$). The preregistered fixed rule A0 used $z(F1) + z(F2) - z(F4) + z(F5)$; the data reject the signs of F4 and F5.

2.2 Data

TABLE 2. Data sources (all free; links in Appendix B).

CONFIRMATORY

Source	Series	Freq.	Known at t	Vintage handling in the frozen study
BLS JOLTS	openings, hires, quits, layoffs rates; 13 industries; not seasonally adjusted	M	$t - 2$	ALFRED vintages: the value known on each decision date
BLS CES	average weekly hours and hourly earnings, production and nonsupervisory workers	M	$t - 1$	as above
FRED	total openings (JTSJOL) and unemployed (UNEMPLOY) for T_t	M	$t - 2$	as above
Ken French	49 value-weighted industry portfolios $\rightarrow$ 13 groups; five factors and momentum	M	t	CRSP build 202608

Universe, a design constraint. The 49 French industries are mapped to the 13 JOLTS/CES industry groups and value-weighted within each group (Appendix C). Thirteen groups bound what any signal can earn: by the fundamental law of active management (Section 3.3) the information ratio is about $IC \times \sqrt{BR} \times TC$, and with 13 groups a monthly book has at most $13 \times 12 = 156$ independent bets a year and a 12-month book 13, so a monthly IC of 0.05 caps the ratio near 0.6 and a 12-month IC of 0.10 near 0.35. The mapping is also coarse: labor data describe every establishment in an industry, returns describe listed firms, and durable manufacturing is 60% semiconductors by market capitalisation.

Point-in-time data. Labor data are revised after first release, so today's values leak information. The frozen study rebuilt every series as it stood on each decision date from ALFRED, the St. Louis Fed archive of data vintages (78 series), and kept three alternatives: first release, fully revised, and revised with fixed publication lags. The primary result barely depends on the choice: A3's test Sharpe is -0.58 as known, -0.52 on first releases, -0.56 revised and -0.53 with fixed lags. Our post-hoc work (Section 3.3) uses revised data with fixed lags, which reproduces the frozen fixed-lag signal with a correlation of 0.998 across all group-months; as-known data pick the same top-three groups in only 42% of months, so post-hoc holdings, though not the conclusions, would have differed in real time.

Sample. Research (signal design and model selection): May 2004 to September 2014, 125 months. Test, opened once: November 2014 to August 2026, 142 months. Figure 2 shows why H3 could not be learned: V/U never exceeded 1 in the research period and exceeded it in 75 test months.

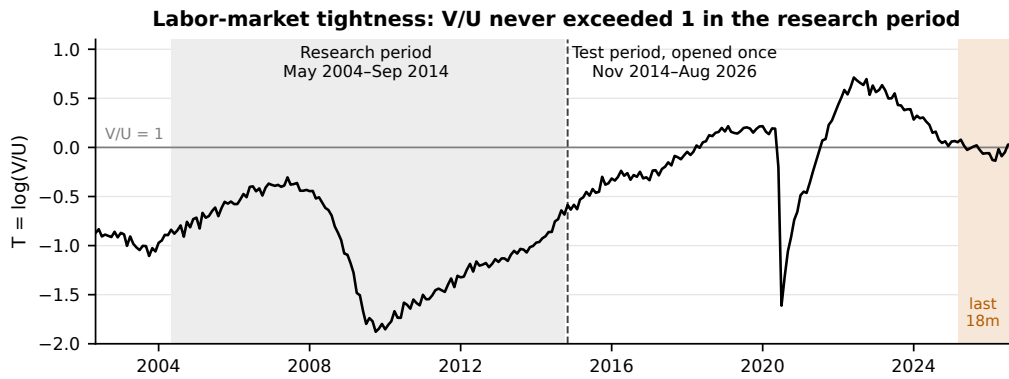


FIGURE 2. Tightness never exceeded $V/U = 1$ in the research period, so a moderator learned there (H3) is extrapolated throughout the test.

POST-HOC

2.3 Implementation

Signals. For a labor series $x_{g,t}$ in group g , let $\bar{x}_{g,t} = \frac{1}{3}(x_{g,t} + x_{g,t-1} + x_{g,t-2})$. With publication lag ℓ (2 months for JOLTS, 1 for CES),

$$F_{g,t} = \bar{x}_{g,t-\ell} - \bar{x}_{g,t-\ell-12} \quad (\text{log differences for hours and earnings}). \quad (1)$$

Each signal is divided by its own expanding-window standard deviation (at least 24 months), z-scored across the 13 groups and winsorised at ± 3 , giving $z_{g,t}^{(k)}$. Year-over-year differences remove seasonality, so the not-seasonally-adjusted series that exist for every industry can be used.

Models. Table 3 lists the five preregistered strategies, all of them H1 books. The learned ones are ridge regressions of next-month relative return on the signals,

$$\hat{\beta}_t = \arg \min_{\beta} \sum_{s < t} \sum_g (r_{g,s+1}^{\text{rel}} - z_{g,s}^{\top} \beta)^2 + \alpha \|\beta\|^2, \quad (2)$$

re-estimated each month on an expanding window, with α chosen once by cross-validation on contiguous time blocks with a one-month purge, inside the research period, over a grid from 0.01 to 1000. Cross-validation chose 1000, the strongest shrinkage the grid allowed, for every model.

TABLE 3. Preregistered strategies (legend for the codes used below).

CONFIRMATORY

A0	Fixed rule, no estimation: $S_{g,t} = \frac{1}{4}(z^{(1)} + z^{(2)} - z^{(4)} + z^{(5)})$.
A1	Ridge on F1–F5.
A1-T	A1 plus each signal interacted with tightness T (H3).
A3	A1 plus features proposed by <i>independent</i> LLM agents (Section 2.3); the primary, chosen on research data.
A4	A1 plus features proposed by <i>communicating</i> LLM agents.

Portfolio, risk model and costs. Each month the score $s_{m,g}$ known at the end of month $m - 1$ is converted to a dollar-neutral tranche (one month’s positions), long the three highest-ranked groups and short the three lowest; the book averages the last three tranches, is hedged to zero market beta, scaled to a 10% ex-ante volatility using a Ledoit–Wolf shrinkage covariance (Ledoit and Wolf, 2004) and capped at a gross exposure of 2, and pays 10 bp per unit of industry turnover plus 2 bp on the hedge:

$$\tau_{m,g} = \begin{cases} +\frac{1}{3} & g \in \text{top 3 of } s_{m,\cdot} \\ -\frac{1}{3} & g \in \text{bottom 3 of } s_{m,\cdot} \\ 0 & \text{otherwise} \end{cases}, \quad b_m = \frac{1}{H} \sum_{j=0}^{H-1} \tau_{m-j}, \quad H = 3, \quad (3)$$

$$h_m = -\sum_g b_{m,g} \hat{\beta}_{m,g}, \quad x_m = (b_{m,1}, \dots, b_{m,13}, h_m), \quad (4)$$

$$k_m = \min\left(\frac{0.10}{\sqrt{12 x_m^\top \hat{\Sigma}_m x_m}}, \frac{2}{\|x_m\|_1}\right), \quad w_m = k_m x_m, \quad (5)$$

$$r_m^{\text{net}} = \sum_g w_{m,g} r_{m,g}^e + w_{m,M} \text{MKT}_m - c \sum_g |w_{m,g} - \tilde{w}_{m,g}| - c_h |w_{m,M} - \tilde{w}_{m,M}|. \quad (6)$$

where $s_{m,g}$ is the score of group g known at the end of month $m - 1$, $\hat{\beta}_{m,g}$ and the Ledoit–Wolf covariance $\hat{\Sigma}_m$ (13 groups plus the market) use the 60 months ending in $m - 2$ (at least 36), $r_{m,g}^e$ is the group’s excess return, $\tilde{w}$ are the previous weights after drift, $c = 10$ bp and $c_h = 2$ bp. In the test the cap, rather than the volatility target, bound in 87% of A3’s months, so realised volatility ran at 7.8% against the 10% target.

HOW WE KEPT THE TEST HONEST

- Every input as it was known on the decision date (ALFRED vintages); the primary result is unchanged under four data vintages.
- Signals, models, the primary strategy and five pass/fail gates (G1 timing, G2 prediction, G3 incremental alpha, G4 robustness, G5 implementability) were fixed and hashed before any test data were touched; the test period was opened exactly once.
- Every result reported whatever its sign: a circular-shift placebo (the signal panel is rotated against returns by every shift from 24 months up, with the model refitted each time, and the real IC is compared with the shifted ones), drop-one-industry checks, bootstrap comparisons of the four registered strategy pairs with a Holm correction for multiple tests, and a deflated Sharpe ratio (DSR), which discounts a Sharpe ratio for the number of trials, or “looks”, taken (Bailey and López de Prado, 2014).
- Anything decided after seeing test results is labelled **POST-HOC** and was itself written to a file and hashed before it ran; the paper book is labelled **FORWARD TEST**. [TODO: Alex: confirm the freeze hashes and the single opening against FREEZE.json and evaluation_started.json]

AI-assisted signal search. We asked large language model (LLM) agents to propose new labor features from six anonymised variables using a restricted formula grammar. Two arms of three agents each, *independent* versus *islands* that exchange short migration reports, ran three repetitions, for 108 proposals per study (Figure 3). Selection was mechanical (research $t \geq 1.5$, same sign in both halves, low correlation, at most three per arm). An LLM judge, audited by a human, scored whether ideas

spread between agents. The agents were Claude Sonnet 5.5 in the confirmatory study and Claude Opus 5.5 in an exploratory follow-up run after the test had been seen (Appendix F).

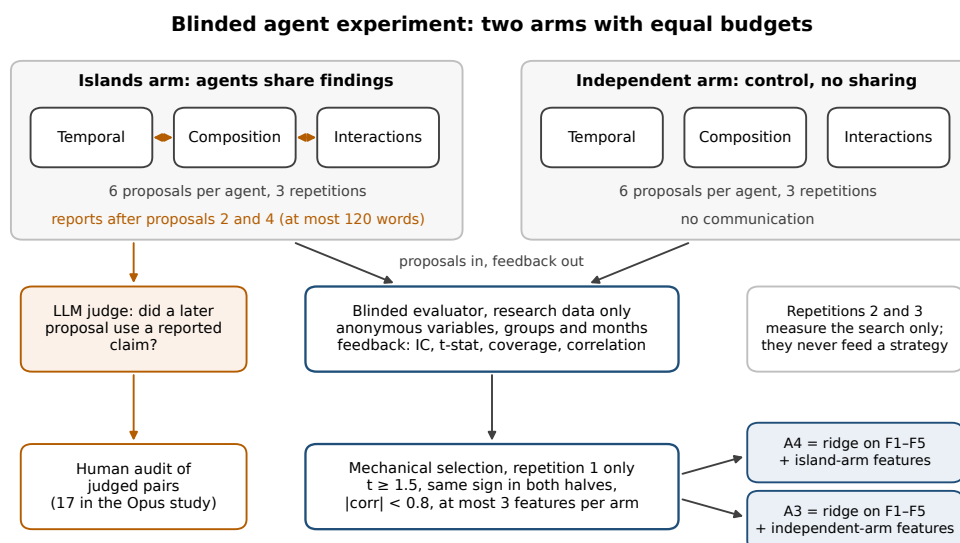


FIGURE 3. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge.

Claude as research assistant. We used Claude (Anthropic) at four points of the build: auditing sources and code (P1), proposing signals as blinded agents (P2), judging whether ideas spread (P3) and red-teaming the robustness plan (P4). We used it again to review the finished study (P5) and to run this iteration (P6). Three of the prompts, verbatim, are below; outputs and our evaluation are in Section 3.4, transcripts in Appendix A.

PROMPT P1: SOURCE AND CODE REVIEW

Find mapping errors, timestamp or look-ahead errors, and code bugs in the material below. For each problem, give the exact line or row, why it is wrong, and a test that would confirm it. Do not report style issues. [followed by the series manifest, the industry crosswalk and the as-of code]

PROMPT P2: BLINDED SIGNAL-SEARCH AGENT (SYSTEM PROMPT, OPENING)

You are a quantitative researcher. You will propose one feature per turn to predict which of 13 anonymous groups (G01 to G13) will have higher returns next month relative to the others. You see only the variable dictionary below, the allowed operations, and summary statistics of your previous submissions. You do not know what the groups, variables or time periods are. Do not guess them and do not use knowledge about any real industry, company or date. [variable dictionary, grammar and JSON format follow]

PROMPT P4: ROBUSTNESS RED TEAM

Give 10 distinct reasons these results could be spurious, ranked by likelihood. For each, name a test that can be run with this data and the result that would make us say "do not implement". [followed by the research-period result tables, without dates or names]

3 What did we learn?

ANSWER TO THE THESIS

- *H1, demand channel: rejected* **CONFIRMATORY**. The primary H1 book lost 4.6% a year (Sharpe -0.58), its IC was -0.021 ($t = -1.08$), the placebo p was 0.72 and four of five gates failed.
- *H2, cost channel: supported, fragile in time* **POST-HOC**. The hiring score has a 12-month IC of $+0.096$ ($t = 2.43$) and a 12-month book is positive in every window ($+0.29$ test), but it lost money in the first half of the test and its deflated Sharpe is 0.05; a forward test is live.
- *H3, tightness: untestable as designed*. V/U never exceeded 1 in the research sample, so 133 of 142 test months fell in one tightness bin.

Table 4 reports the one-time test of H1; no strategy made money, and costs are not the explanation, since A3 paid 1.1% a year in trading costs against a gross loss of 3.5%.

TABLE 4. Preregistered H1 strategies: research versus the one-time test.

CONFIRMATORY

Strategy	Research	Test, Nov 2014–Aug 2026 (142 months)			
	Sharpe	Sharpe	Net return	Volatility	Max drawdown
A0 fixed rule	-0.44	-0.36	-3.5%	9.7%	-40.3%
A1 ridge	-0.54	-0.51	-4.0%	7.9%	-37.7%
A1-T tightness	-0.55	-0.59	-4.9%	8.2%	-36.3%
A3 primary	-0.13	-0.58	-4.6%	7.8%	-41.1%
A4 islands	-0.18	-0.79	-6.1%	7.8%	-53.3%

Annualised, net of costs, as-known data; research Sharpe over the 88 months in which every model had a forecast. Primary: factor alpha -0.38% a month ($t = -1.86$), Sharpe -0.79 at 25 bp costs, 41% maximum drawdown; every Holm-adjusted pairwise comparison had adjusted $p = 1.0$.

3.1 Style and risk exposures

The H1 books are quality and growth tilts rather than bets on labor news. Regressed on the Fama–French five factors, momentum and an industry-momentum factor (Figure 7, appendix), the primary loads on profitability (RMW $+0.19$, $t = 2.2$) and against value (HML -0.15 , $t = -2.0$), with an alpha of -0.38% a month ($t = -1.86$). That is plausible: industries adding workers and hours are the profitable, growing ones, so a long-hiring book buys profitability and sells value. The post-hoc wage-growth book tilts the same way but harder (HML -0.43 , $t = -4.4$; momentum $+0.22$, $t = 3.3$), while the H2 book loads the other way, against profitability (RMW -0.36 , $t = -4.45$).

3.2 Time patterns of performance

The H1 primary lost money in every window except the last 18 months, and that bright spot ($+0.85$) rests on an IC t of 0.44 (Table 5, Figure 4). For A3, “post-2010” is in effect the test, which starts in November 2014; “full” adds the research years and is descriptive. The post-hoc books show the opposite pattern: wage growth did well until 2024 and lost heavily in the last 18 months (-1.71), while the post-hoc H2 book is the only one positive in every window.

TABLE 5. Net Sharpe ratio by window.

CONFIRMATORY (A3) / POST-HOC (V2a, V2d)

Window (returns)	Research 2004–14	Test 2014–26	Full sample 2004–26	Post-2010 2010–26	Last 18 months 2025-03–2026-08
A3 primary, H1 (confirmatory)	−0.13	−0.58	−0.44	−0.62	+0.85
V2a wage growth (post-hoc)	+0.09	+0.16	+0.15	+0.12	−1.71
V2d −(hires+quits), 12m, H2 (post-hoc)	+0.50	+0.29	+0.35	+0.30	+0.29

A3's research and full-sample figures use the 88 months with a forecast (from June 2007) and are descriptive; V2a and V2d use revised data with fixed lags over 125, 142, 268, 200 and 18 months.

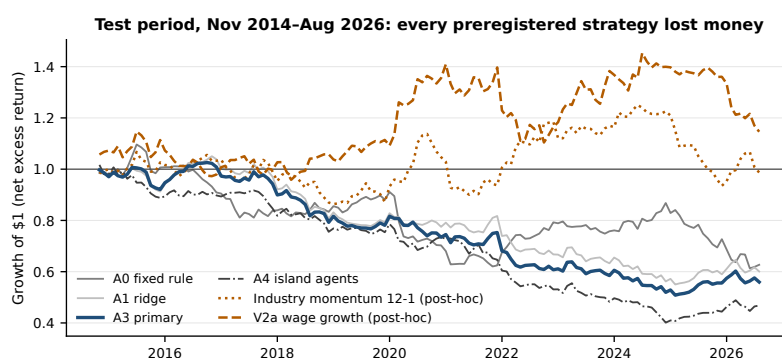


FIGURE 4. Every preregistered H1 strategy lost money in the test; the post-hoc wage-growth book gained until 2024 and gave much of it back.

CONFIRMATORY (A0–A4) / POST-HOC (momentum, V2a)

3.3 Robustness and limits

What the universe allows. The fundamental law sizes every result. In the test, A3's IC of -0.021 and transfer coefficient (TC) of 0.80 imply an IR of -0.21 against a realised -0.58 ; V2a's IC of $+0.049$ implies $+0.56$ against $+0.16$; the H2 book's 12-month IC of $+0.096$ with a TC of 0.48 implies $+0.17$ against $+0.29$. A modest Sharpe ratio is the ceiling here, and the preregistered overlay (hedge, volatility target, binding cap, 0.9 turnover a month) cost the monthly books most of what their IC allowed.

Why H1 failed. Four post-hoc explanations (figures in Appendix D).

1. *Wrong signs.* Layoffs carry a positive IC in both periods and hours a negative one, the opposite of the signs A0 imposed (Table 1); wage growth, dropped during the build, is the only signal with a positive IC in both periods ($+0.029$ research, $+0.049$ test).
2. *Wrong horizon.* At one month no hiring signal predicts returns; at twelve months hires (-0.062 , $t = -2.1$) and quits (-0.065 , $t = -1.7$) carry the H2 sign.
3. *An untestable moderator (H3).* Tightness terciles were set on an expanding window, so 133 of 142 test months fell in the “high” tercile; a split on whether T rose or fell over 12 months (93/32 research months, 88/54 test) would have been testable.
4. *Fragile agent features.* The two features selected for the primary carried a regime gate (switched on only in the “low tightness” regime) and were active in 3 of 142 test months; every selected feature, in both studies, failed the frozen test criterion of $t \geq 1.5$ with a positive IC in both halves.

Losses were concentrated: health care cost the primary 26 points of cumulative return, construction 19 and finance 16, against 13 points of trading costs.

What survives (H2), and how much. None of the four post-hoc variants meets the pre-declared reading rule (positive Sharpe at 25 bp, IC $t \geq 2$, alpha $t \geq 2$, positive IC in both test halves, placebo $p < 0.05$, DSR above 0.95). We froze them in a hashed file (`analysis/v2_spec.md`) after the test was opened, ran each once and report all four (Table 6, Figure 5). The declared headline, a wage-growth

rank (V2a), has the right IC (+0.049, $t = 2.04$, placebo $p = 0.02$) but no alpha ($t = 0.29$) and a DSR of 0.02; adding momentum (V2c) raises the Sharpe to +0.35 because momentum supplies the return.

V2d is the H2 book: $-\frac{1}{2}(z(F2) + z(F3))$, short the industries that hire and lose workers fastest, held twelve months. Its score has no one-month IC (-0.020 , $t = -0.87$) and a positive IC from 9 to 24 months, peaking at 12 (+0.096, $t = 2.43$; Figure 6); its alpha is +0.26% a month ($t = 1.48$ test, 1.92 full sample, 12 Newey–West lags; Table 6 uses 6), and it fails the rule.

TABLE 6. Post-hoc variants frozen in `analysis/v2_spec.md`: net Sharpe by window and test statistics at 10 bp. IC, alpha t and placebo are one-month statistics with 6 Newey–West lags; DSR counts 112 looks.

POST-HOC

Variant	Research	Test	Full	Post-2010	Last 18m	Test IC (t)	Alpha t	Placebo p	DSR
V2a	+0.09	+0.16	+0.15	+0.12	-1.71	+0.049 (+2.04)	+0.29	0.02	0.02
V2b	-0.06	-0.15	-0.11	-0.16	-1.34	+0.042 (+1.65)	-0.84	0.05	0.00
V2c	+0.27	+0.35	+0.32	+0.24	-0.78	+0.032 (+1.21)	+1.36	0.26	0.09
V2d	+0.50	+0.29	+0.35	+0.30	+0.29	-0.020 (-0.87)	+1.37	0.85	0.06

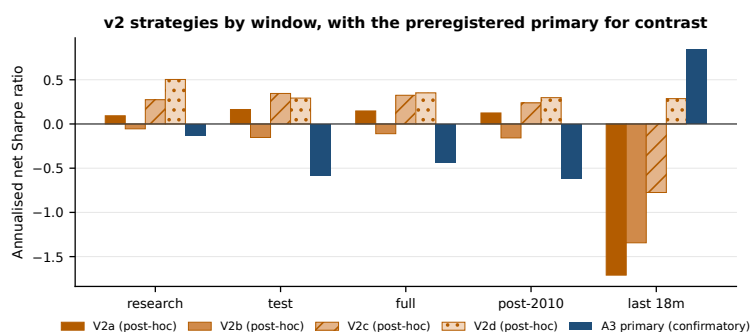


FIGURE 5. V2d is the only post-hoc variant positive in every window; V2a and V2c lost heavily in the last 18 months.

POST-HOC / CONFIRMATORY (A3 bars)

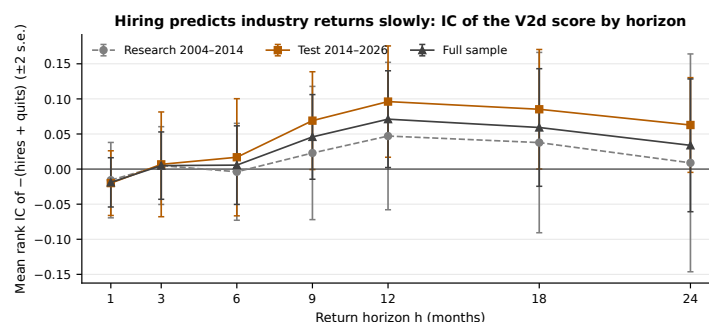


FIGURE 6. Post-hoc horizon profile of the H2 score: no one-month IC and a positive IC from 9 to 24 months, peaking at 12 (test +0.096, $t = 2.43$).

POST-HOC

How robust is the H2 book? It fails one of the seven pre-declared checks and passes the other six (Table 7). We froze the checks in `analysis/v3_spec.md`, raised the look count to 132 and ran them once; after 132 looks the DSR is 0.05 in the test and 0.16 over the full sample, so we read V2d as a hypothesis consistent with the economics and short of evidence. Its behaviour across tightness regimes is recorded without a claim in Figure 12.

TABLE 7. V2d robustness battery, test window unless stated.

POST-HOC

Check	Result	Holds?
Components, held 12 months	hires alone +0.19, quits alone +0.16	yes
Holding period	Sharpe +0.14 (9 months), +0.29 (12), +0.30 (18)	yes
Both test halves positive	first half −0.21, second half +0.64; 12-month IC positive in both (+0.097, +0.095)	no
Costs	break-even 68 bp; +0.22 at 25 bp, +0.09 at 50	yes
Drop one industry	test Sharpe between +0.18 (without health care) and +0.48 (without real estate)	yes
Block bootstrap (12-month blocks, 5,000 draws)	90% interval [−0.07, +0.65] test; [+0.05, +0.64] full sample	yes (full)
Placebo	one-month $p = 0.85$ (Table 6); horizon-matched $p = 0.06$ test, 0.04 full, on overlapping returns, added after the run	outside rule

Forward test. Four paper books with no capital behind them were frozen on 3 October 2026 (hash 735a8ffa, amended by c76cde86 after a data-release check stopped the first run) for the 30 September decision, from data published by 29 September; each has three longs, three shorts and a SPY beta hedge, held to 31 December (Appendix G). V2c and V2d were added after the historical v2 results had been seen, a selection step recorded as one. One quarter has almost no power: a Q4 return between −10% and +10% is consistent with no skill.

Limits. With 142 test months only a Sharpe near 0.6 reaches $t \approx 2$, so a null does not rule out a smaller effect, and none of the post-hoc Sharpe ratios is distinguishable from zero after deflation. The 49-to-13 mapping dilutes the labor signal; the one near-positive result in the Opus follow-up came from a single long position in a group that is 60% semiconductors (Appendix F). All post-hoc numbers use revised data with fixed lags; the as-known rerun was not run for lack of an API key. Stock-level baskets by NAICS code from CRSP are the follow-up we would run next.

3.4 Claude evaluation

Claude was the AI research assistant, which this term’s version of the course allows. Table 8 records every interaction. The models are taken from the call logs: Sonnet 5.5 at medium effort for the 199 logged calls of the confirmatory study, Opus 5.5 at extra-high effort for the 294 calls of the follow-up. [TODO: Alex: exact name and vendor of the engineering assistant that wrote the pipeline code (“6 astra”); it is not documented in the repository]

Agents: Opus against Sonnet. A stronger model followed the protocol better and found no more signal: more valid proposals (106 of 108 against 98), more admitted migration reports (15 of 18 against 7), far more usable judge labels (84 of 88 against 8 of 36) and more features past the research filter (7 against 4), yet no selected feature survived the test criterion in either study. Communication raised research search scores for Opus (best IC 0.136 against 0.059) and lowered them for Sonnet (0.049 against 0.083); neither gap exceeded the spread across repetitions.

Reflection. **draft** Claude was most useful where the task had a checkable answer (reproducing numbers, auditing code and timing) and least useful where it needed judgement about evidence (features that were artefacts of a regime gate, a judge that over-attributed, a red team that invented statistics). What protected us was mechanics: blinding, a fixed evaluator, hashed specifications and a test opened once. [TODO: Al Yazid: rewrite in the team’s words; remove the draft mark]

TABLE 8. Claude as research assistant: six interactions (what we asked, in the first column; prompts in Section 2.3).

CONFIRMATORY (P1–P4) / POST-HOC (P5–P6)				
Interaction	What it got right	What it got wrong	What we changed	Our critique
P1 audit: find mapping, timing and code errors in the manifest, crosswalk and as-of code	25 findings; the shared CES series for groups 9 and 10	Several alleged look-ahead errors were not errors	Mapping metadata relabelled; no basket changed	[TODO: critique: Alex]
P2 agents: propose features from six anonymous variables within a grammar	Valid grammar in 98 of 108 proposals (106 for Opus); both studies rediscovered wage growth blind	The best research t -statistics came from regime gates active in 3 test months; every selected feature failed the frozen test criterion	Selection stayed mechanical; v2 restores wage growth as a public, ungated signal	[TODO: critique: Elouan]
P3 judge: write migration reports; judge whether later proposals used a reported claim	Opus: 15 of 18 reports admitted, 84 of 88 labels usable	Sonnet: 28 of 36 judge outputs broke the JSON-only rule; the judge said yes 10 times where the human said 3, all 7 disagreements judge-yes	Judge labels reported as associations next to the human labels	[TODO: critique: Al Yazid]
P4 red team: ten reasons the research results could be spurious, each with a test	Ranked, specific; called “do not implement” from research tables before the seal opened	Quoted p-values it had not computed; proposed unregistered thresholds	No new test adopted; each concern was already covered	[TODO: critique: Romain]
P5 review: read the frozen study and say what is wrong with it	Wrong signs, dropped wage signal, 12-month hiring sign, 3-month agent features, one-bin tightness split	Guessed the engineering assistant’s vendor from a path in the code; a momentum definition that included the latest month; wage ICs with a non-frozen start date	Momentum redefined; wage ICs recomputed; the guess withdrawn with a correction note	[TODO: critique: Piero]
P6 iteration: rebuild the analysis, freeze and run v2/v3, build the forward book, draft this report	Reproduced the sealed returns to 10^{-16} ; froze every spec before running; stopped at a failed gate	Assumed August JOLTS would not be out by 29 September; it was, so its own check stopped the run	Amendment 001, disclosed; run logs that keep every result reproducible	[TODO: critique: Alex]

References

- David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- Frederico Belo, Xiaoji Lin, and Santiago Bzdresch. Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177, 2014.
- Andres Donangelo, François Gourio, Matthias Kehrig, and Miguel Palacios. The cross-section of labor leverage and equity returns. *Journal of Financial Economics*, 132(2):497–518, 2019.
- Harrison Hong, Walter Torous, and Rossen Valkanov. Do industries lead stock markets? *Journal of Financial Economics*, 83(2):367–396, 2007.
- Lars-Alexander Kuehn, Mikhail Simutin, and Jessie Jiaxu Wang. A labor capital asset pricing model. *Journal of Finance*, 72(5):2295–2332, 2017.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- Tobias J. Moskowitz and Mark Grinblatt. Do industries explain momentum? *Journal of Finance*, 54(4):1249–1290, 1999.
- Carlo Zarattini and Gary Antonacci. A century of profitable industry trends. SSRN Working Paper 4857230, 2024.

A Claude interaction transcripts

Paths are relative to 230ga-follow-the-workers/outputs/follow_the_workers/ unless stated; the Opus follow-up has the same files under follow_the_workers_opus55_xhigh/.

- P1: prompt outputs/p1_request.json (the instruction in Section 2.3 followed by the manifest, crosswalk and as-of code); response outputs/p1_response.md; dispositions outputs/p1_review.md. [TODO: Alex: add the disposition counts from p1_review.md]
- P2: system and turn prompts in params.py (SYSTEM_PROMPT, TURN_PROMPT); every call with its raw response in outputs/agents/log.jsonl; proposals in outputs/candidates.csv; selected features decoded in analysis/output/tables/d7_selected_features.csv.
- P3: migration reports in outputs/agents/migrations.json; the judge prompt in agents.py, function trace_migrations; judgments in outputs/agents/migration_judgments.json; the human audit in outputs/human_review_submission.json (the Opus folder also has a .md version).
- P4: prompt outputs/p4_request.json; response outputs/p4_response.md; dispositions outputs/p4_review.md.
- P5: the review and its output, FINDINGS_AND_PROPOSAL.md, with the numbers in analysis/output/diagnostics.md.
- P6: the task files ITERATION_PROMPT.md and OVERNIGHT_PROMPT.md; the rules in CLAUDE.md; outputs in analysis/run.ipynb, analysis/output/, results/ and this report.

B Data links

- Ken French Data Library: https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html (49 industry portfolios, five factors, momentum; CRSP build 202608).
- JOLTS by industry, not seasonally adjusted: FRED series JTU<code>JOR/HIR/QUR/LDR, for example <https://fred.stlouisfed.org/series/JTU2300HIR>; codes 110099, 2300, 3200, 3400, 4200, 4400, 480099, 5100, 5200, 5300, 540099, 6200, 7200.
- CES hours and earnings: CEU<code>07 and CEU<code>08 (construction earnings CES2000000008); codes 10000000, 20000000, 31000000, 32000000, 41420000, 42000000, 43000000, 50000000, 55000000, 60000000, 65000000, 70000000.
- Tightness: JTSJOL and UNEMPLOY. Vintages: <https://alfred.stlouisfed.org/>.
- ETF proxies and their caveats: analysis/forward/etf_proxies.csv. [TODO: Elouan: check every link and add access dates]

C Industry mapping

TABLE 9. The 13 groups: JOLTS code, Ken French members and the largest member's share of August 2026 market capitalisation.

CONFIRMATORY

Group	JOLTS	Ken French industries	Largest member
Mining and logging	110099	Gold, Mines, Coal, Oil	Oil 79%
Construction	2300	Cnstr	100%
Durable manufacturing	3200	Toys, BldMt, Steel, FabPr, Mach, ElcEq, Autos, Aero, Ships, Guns, Hardw, Chips, LabEq, MedEq	Chips 60%
Nondurable manufacturing	3400	Food, Soda, Beer, Smoke, Hshld, Clths, Drugs, Chems, Rubbr, Txtls, Paper	Drugs 60%
Wholesale	4200	Whsl	100%
Retail	4400	Rtail	100%
Transport and utilities	480099	Trans, Util	Util 61%
Information	5100	Telcm	100%
Finance and insurance	5200	Banks, Insur, Fin	Banks 51%
Real estate	5300	REst	100%
Professional and business services	540099	BusSv	100%
Health care	6200	Hlth	100%
Accommodation and food	7200	Meals	100%

Groups 9 and 10 share one CES series, and groups 7, 12 and 13 use broader CES supersectors, so the effective cross-sectional width is closer to 8 or 9 than 13.

D Full results and figures

Data vintages. A3 test Sharpe: as-known -0.58 , first release -0.52 , revised -0.56 , fixed lags -0.53 ; IC -0.021 , -0.040 , -0.026 , -0.021 .

Cost sensitivity. A3 Sharpe -0.45 at 0 bp and -0.79 at 25 bp industry costs. V2d: $+0.34$, $+0.29$, $+0.22$ and $+0.09$ at 0, 10, 25 and 50 bp; break-even 68 bp.

Gates. G1 timing: pass. G2 prediction (IC $t \geq 2$): fail. G3 incremental alpha ($t \geq 2$): fail. G4 robustness (positive IC in both halves, placebo $p < 0.05$, no industry above half the profit): fail. G5 implementable (Sharpe > 0 at 25 bp, capacity documented): fail.

Primary by window. First half -0.72 , second half -0.49 , excluding March–December 2020 -0.51 , last 18 months $+0.85$.

Where the numbers live. Signal, horizon, tightness, risk-overlay and agent tables: `analysis/output/diagnostics_v2.md`; v2 statistics: `analysis/output/v2_results.md`; the robustness battery: `analysis/output/v3_results.md`; every claim with its source: `results/claims.csv`.

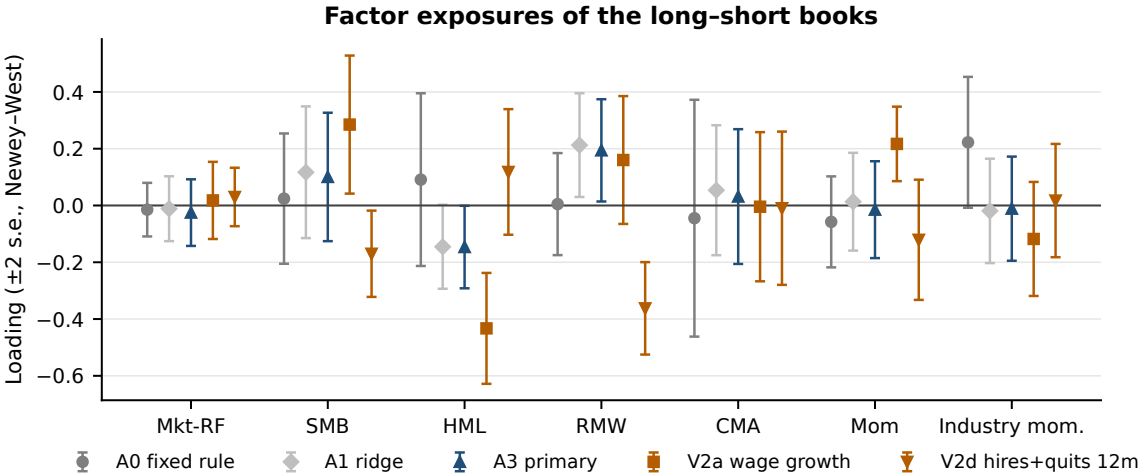


FIGURE 7. Factor loadings of the long-short books: A3 buys profitability and sells value; V2a sells value and buys momentum; V2d sells profitability. CONFIRMATORY (A0, A1, A3) / POST-HOC (V2a, V2d)

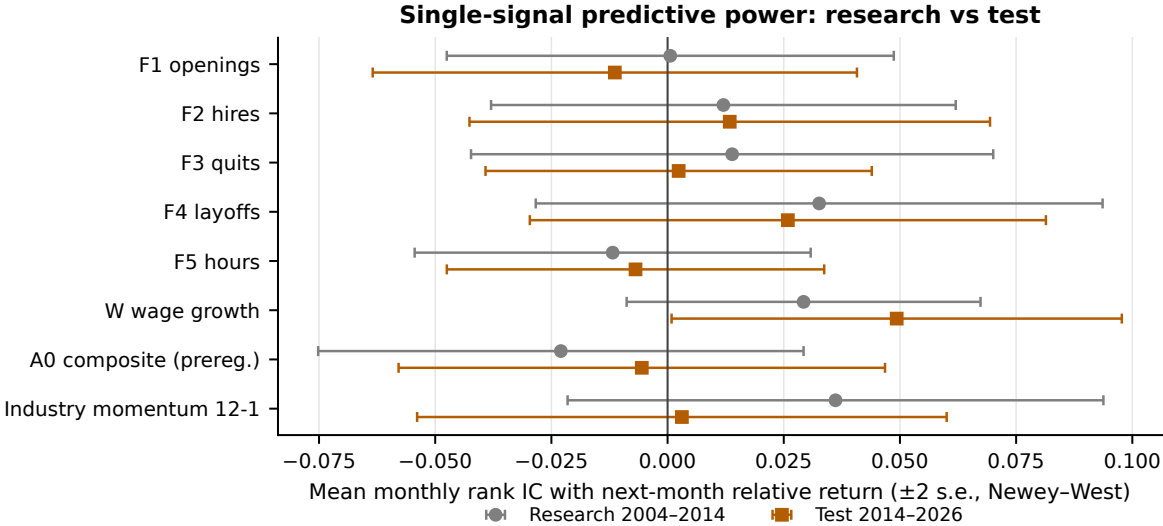


FIGURE 8. Only wage growth has a positive IC in both periods; layoffs and hours carry the opposite sign to the one the fixed rule imposed. POST-HOC

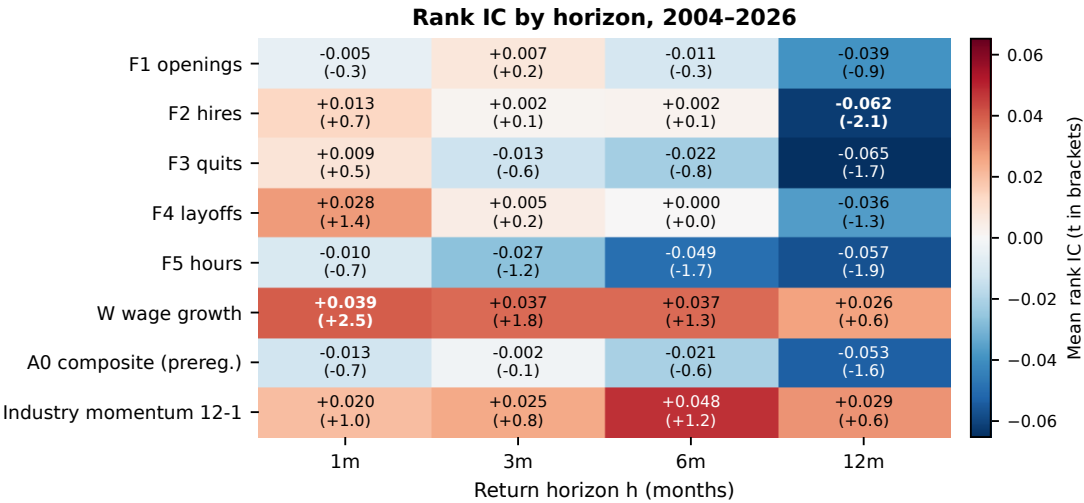


FIGURE 9. Hires and quits turn negative at 12 months ($t = -2.1, -1.7$): the investment channel shows up slowly, not within a month. POST-HOC

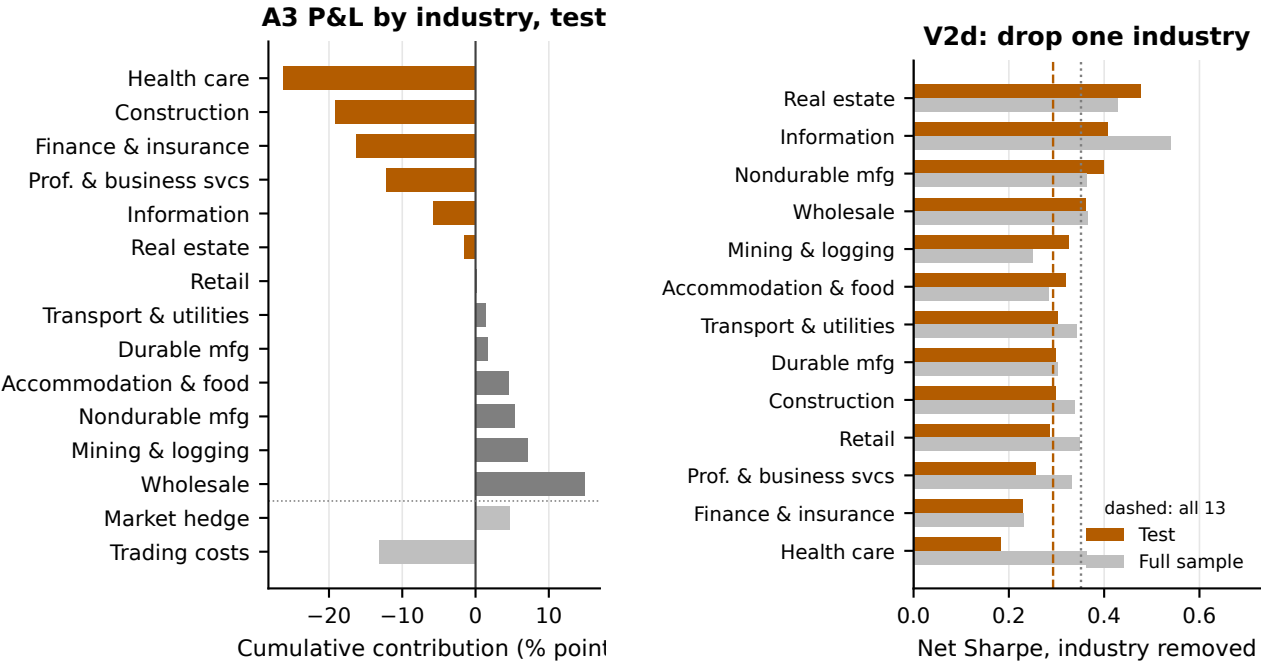


FIGURE 10. Left: the primary lost most in health care, construction and finance. Right: no single industry carries V2d; removing any one leaves its test Sharpe between +0.18 and +0.48.

POST-HOC

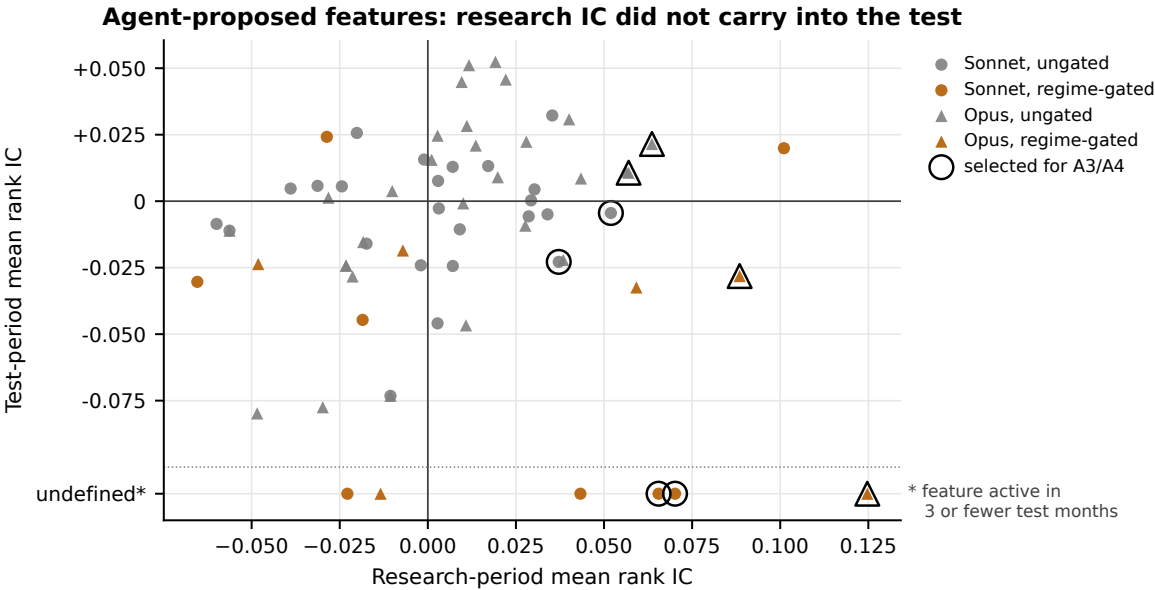


FIGURE 11. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months.

POST-HOC

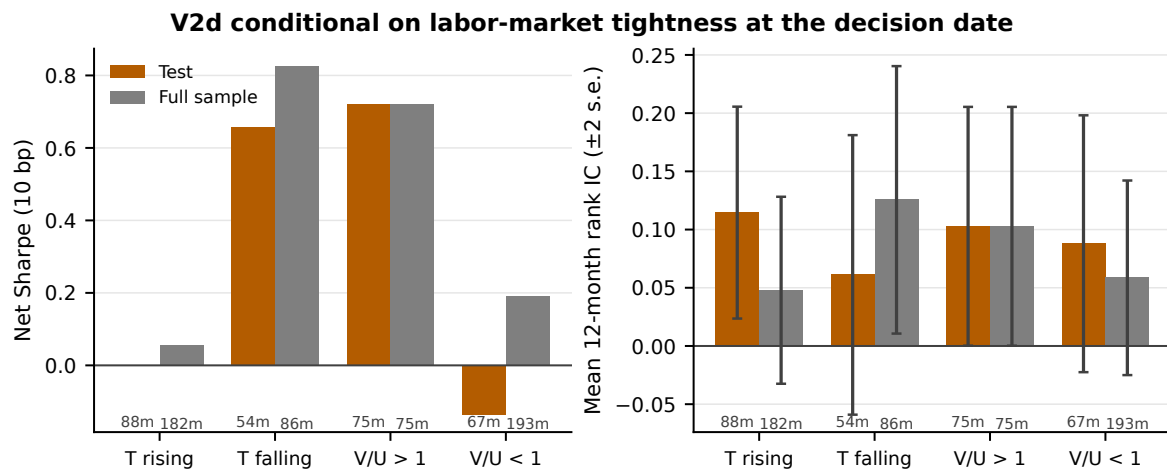


FIGURE 12. V2d conditional on tightness at the decision: it earned its return when tightness was falling or $V/U > 1$; the pattern rests on 54 to 88 months per cell.

POST-HOC

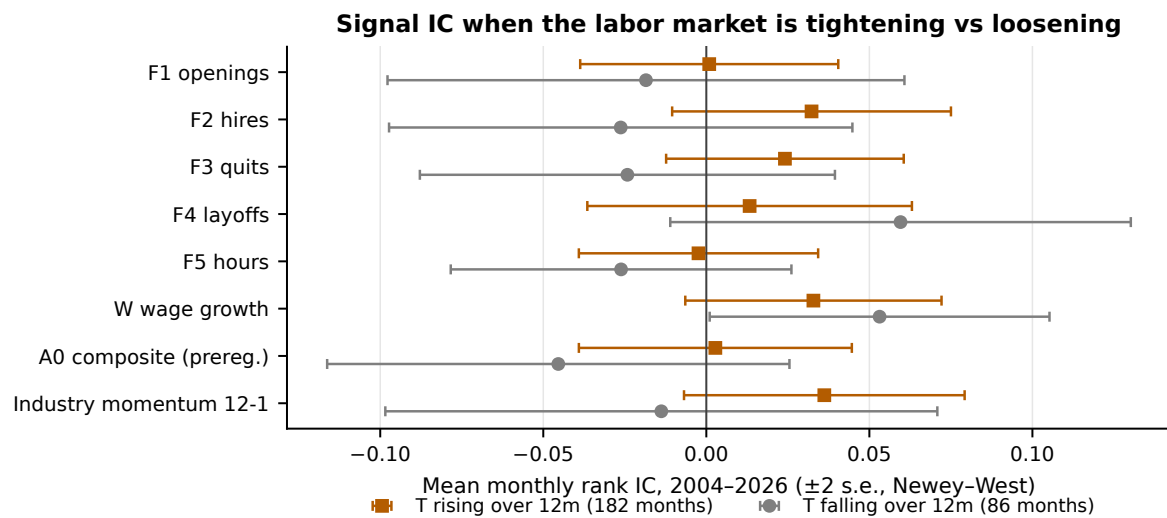


FIGURE 13. Single-signal IC in months when T rose versus fell over the previous 12 months; the split is balanced in both periods.

POST-HOC

TABLE 10. Rank IC by horizon, May 2004–August 2026 (t in brackets).

POST-HOC

Signal	1m	3m	6m	12m
F1 openings	−0.005 (−0.31)	+0.007 (+0.23)	−0.011 (−0.28)	−0.039 (−0.92)
F2 hires	+0.013 (+0.70)	+0.002 (+0.07)	+0.002 (+0.08)	−0.062 (−2.08)
F3 quits	+0.009 (+0.50)	−0.013 (−0.55)	−0.022 (−0.78)	−0.065 (−1.73)
F4 layoffs	+0.028 (+1.37)	+0.005 (+0.17)	+0.000 (+0.01)	−0.036 (−1.28)
F5 hours	−0.010 (−0.67)	−0.027 (−1.24)	−0.049 (−1.69)	−0.057 (−1.86)
W wage growth	+0.039 (+2.51)	+0.037 (+1.77)	+0.037 (+1.28)	+0.026 (+0.63)
A0 composite (prereg.)	−0.013 (−0.69)	−0.002 (−0.09)	−0.021 (−0.61)	−0.053 (−1.58)
Industry momentum 12-1	+0.020 (+0.99)	+0.025 (+0.80)	+0.048 (+1.24)	+0.029 (+0.63)

TABLE 11. V2d robustness battery: holding period (R3) and sub-periods (R4).

POST-HOC

Holding	research	test	full	post-2010	last 18m	Test turnover
6 months	+0.31	+0.05	+0.15	+0.05	+0.50	0.49
9 months	+0.43	+0.14	+0.24	+0.16	+0.29	0.40
12 months	+0.50	+0.29	+0.35	+0.30	+0.29	0.37
18 months	+0.58	+0.30	+0.38	+0.34	+0.31	0.28

Window	Months	Sharpe	Net p.a.	IC 12m (t)
test: first half	71	-0.21	-1.4%	+0.097 (+1.75)
test: second half	71	+0.64	+6.5%	+0.095 (+1.60)
test: excl. Mar 2020-Dec 2021	120	+0.37	+3.0%	+0.105 (+2.23)
test: pre-2020	62	+0.14	+0.8%	+0.114 (+1.92)
test: 2020 onward	80	+0.37	+3.9%	+0.080 (+1.54)
full: first half	134	+0.33	+2.0%	+0.038 (+0.73)
full: second half	134	+0.38	+3.3%	+0.107 (+2.66)
full: excl. Mar 2020-Dec 2021	246	+0.40	+2.9%	+0.073 (+1.93)
full: pre-2020	188	+0.36	+2.2%	+0.068 (+1.57)

E Code snippets

As-of lookup. The value of a series known on a cutoff date (frozen `data.py`).

```
def vintage_view(frame, cutoff, first=False):
    cutoff = pd.Timestamp(cutoff).strftime('%Y-%m-%d')
    if first:
        eligible = frame.drop_duplicates('observation_month', keep='first')
        eligible = eligible.loc[eligible.realtime_start <= cutoff]
    else:
        eligible = frame.loc[(frame.realtime_start <= cutoff) & (frame.realtime_end >= cutoff)]
    if eligible.duplicated('observation_month').any():
        raise ValueError('Overlapping vintage intervals')
    return eligible.set_index('observation_month')[['value', 'realtime_start']].sort_index()
```

Signal construction. Fixed-lag signals and tightness (`analysis/data.py`).

```
def signals(folder=P.FRED, last_decision=None, groups=None):
    panel = labor_panel(folder)
    avg3 = lambda x: x.rolling(3, min_periods=3).mean()
    raw = {}
    to_decision = lambda frame, lag: frame.set_axis(frame.index + lag) # observation o ->
    decision o+lag
    for i, m in enumerate(P.JOLTS_MEASURES, 1):
        x = avg3(panel[m])
        raw[f'F{i}'] = to_decision(x - x.shift(12), P.JOLTS_LAG)
    h, e = avg3(panel['H']), avg3(panel['E'])
    raw['F5'] = to_decision(np.log(h / h.shift(12)), P.CES_LAG)
    raw['W'] = to_decision(np.log(e / e.shift(12)), P.CES_LAG)
    # last decision month at which every feature has its lagged observation
    end = last_decision or min(v.dropna(how='all').index.max() for v in raw.values())
    decisions = pd.period_range(P.SIGNAL_START, end, freq='M')
    features = {}
    for name, value in raw.items():
        value = value[list(groups)] if groups is not None else value # drop-one-industry
    diagnostics
        z = standardize(value.reindex(decisions))
        z.index = z.index + 1 # earning month
        features[name] = z
    # T at decision d uses the latest month <= d-2 with both V and U observed (frozen rule); this
    # matters once: the Oct 2025 household survey (UNEMPLOY) was never collected.
    ratio = np.log(panel['V'] / panel['U']).ffill()
```

```

t = to_decision(ratio, P.JOLTS_LAG).reindex(decisions)
t.index = t.index + 1
return features, t

```

Backtest core. Tranche, hedge, scaling and costs (frozen data.py, excerpt).

```

for month,score in scores.iterrows():
    tranche = pd.Series(0.,index=groups)
    if signal_available.loc[month]:
        rank = score.sort_index().sort_values(kind='stable')
        tranche.loc[rank.index[-P.PORTFOLIO['longs']:]] = 1/P.PORTFOLIO['longs']
        tranche.loc[rank.index[:P.PORTFOLIO['shorts']]] = -1/P.PORTFOLIO['shorts']
    tranches.append(tranche)
    book = sum(tranches[-holding:])/holding
    betas = pd.Series({g:np.cov(historical[g],market,ddof=1)[0,1]/variance for g in groups})
    raw = np.r_[book.to_numpy(),-float(book@betas)]
    covariance = LedoitWolf().fit(historical.to_numpy()).covariance_
    risk = np.sqrt(max(0,float(raw@covariance@raw))*12)
    if risk<=0 and np.abs(raw).sum(>1e-14:
        raise ValueError('Degenerate portfolio risk estimate')
    scale = min(P.PORTFOLIO['vol_target']/risk,P.PORTFOLIO['gross_cap']/np.abs(raw).sum()) if
risk>0 else 0.
    weights = pd.Series(raw*scale,index=pretrade.index)
    assert abs(weights[groups]@betas+weights['market'])<1e-10
    assert weights.abs().sum(<=P.PORTFOLIO['gross_cap']+1e-10
    turnover = float((weights[groups]-pretrade[groups]).abs().sum())
    hedge_turnover = abs(weights['market']-pretrade['market'])
    transaction = turnover*cost_bps/10000+hedge_turnover*P.PORTFOLIO['hedge_cost_bps']/10000
    # Charge borrowed securities plus negative residual cash, at the stated annual rate.
    borrow_base = -weights.clip(upper=0).sum()
    finance_base = max(0,float(weights.sum()-1))
    financing = (borrow_base+finance_base)*borrow_bps/10000/12
    gross = float(weights[groups]@returns['excess'].loc[month,groups]+weights['market']*factors
['Mkt-RF'])
    net = gross-transaction-financing

```

F Follow-up study with a stronger model

After the test had been opened, the agent experiment was rerun with Claude Opus 5.5 at extra-high effort, with fresh agents and the same frozen evaluator, selection rule and portfolio. Because the test period was already known to the team, the run is exploratory. Its research-selected primary, A4, earned a test Sharpe of +0.01 (research −0.40) with an IC of −0.002 ($t = -0.10$), an alpha t of 0.05, a placebo p of 0.56 and a Sharpe of −0.19 at 25 bp; it failed gates G2 to G5 and had a Sharpe of −1.41 over the last 18 months. Its gross return came from one long position: durable manufacturing contributed +0.27 of cumulative gross return (sum of monthly contributions, in units of the book) in a group that is 60% semiconductors. Opus's other strategy, A3, earned −0.46. The follow-up is evidence about agents (Section 3.4), not about the strategy.

G Forward-test specification

- analysis/v2_spec.md: four post-hoc variants and a reading rule; SHA-256 ca658234..., frozen 2026-10-03T09:00:13Z, first run 09:01:55Z.
- analysis/forward/spec_frozen.md: decision 30 September 2026, information cutoff 29 September, one tranche held to 31 December, 60-month beta hedge in SPY, 10 bp and 2 bp costs; SHA-256 735a8ffa..., frozen 09:03:56Z.
- analysis/forward/spec_amendment_001.md: the fixed-lag information set (JOLTS July 2026,

CES August 2026, as published on 29 September) after the pre-committed check stopped the first run; V2c and V2d added after the historical v2 results were seen and before any Q4 return beyond 1–2 October; SHA-256 `c76cde86...`, frozen 09:16:11Z; book built 09:17:55Z.

- `analysis/v3_spec.md`: the robustness battery; SHA-256 `312aef1d...`, frozen 09:52:27Z, first run 09:53:32Z.
- Books: `analysis/forward/book_2026Q4.csv` (group weights for V2a–V2d), `book_2026Q4_etf.csv` (netted ETF weights), `book_2026Q4_hedge.csv` (SPY hedge per unit: -0.40 , -0.06 , -0.24 , -0.02), `etf_proxies.csv`. Holdings: V2a is long construction, durable manufacturing and information and short mining, health care and accommodation; V2b is long durable manufacturing, retail and transport and short real estate, health care and accommodation; V2c is long construction, durable manufacturing and finance and short retail, transport and professional services; V2d is long finance, real estate and accommodation and short mining, durable manufacturing and retail. V2c’s tradable book has four legs, because its long durable-manufacturing and short professional-services positions both map to the industrials ETF and cancel.
- Power: V2d’s test volatility is about 9% a year, so one quarter’s return has a standard deviation near 5%; a Q4 return between -10% and $+10\%$ is within two standard deviations of zero. The book was first built at 09:17:55Z on 3 October: commit `9f4a50f` (09:18:29Z) holds a byte-identical `book_2026Q4.csv` (SHA-256 `857275...`); `book_first_build.sha256` records that first build and later notebook runs must reproduce it.
- Evaluation: the ETF books from the 30 September close to the 31 December close after costs, and the same books on Ken French group returns once published; reported whatever they show. [TODO: Elouan: weekly tracker (deferred by team decision)]